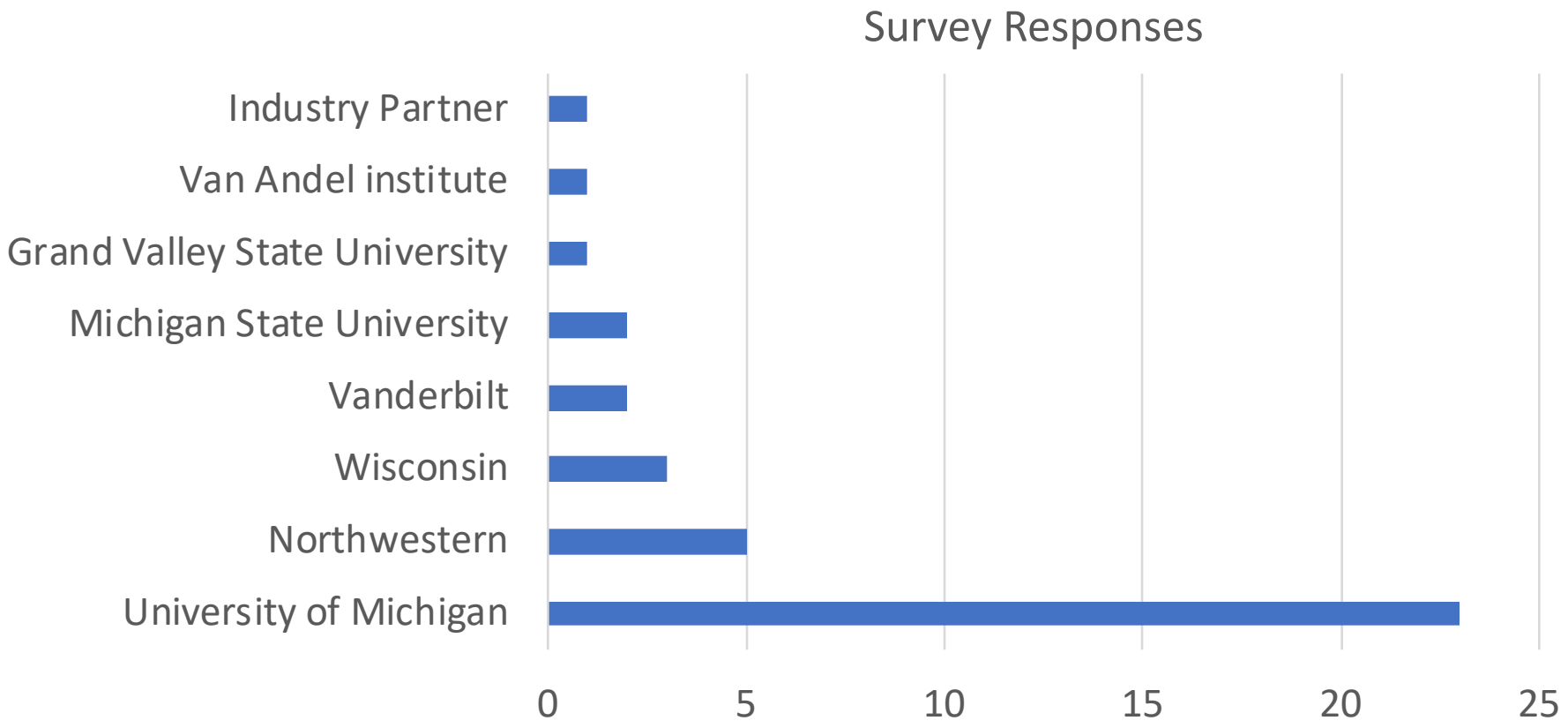
Survey of LS-CAT Users
Future Directions

May 21, 2019

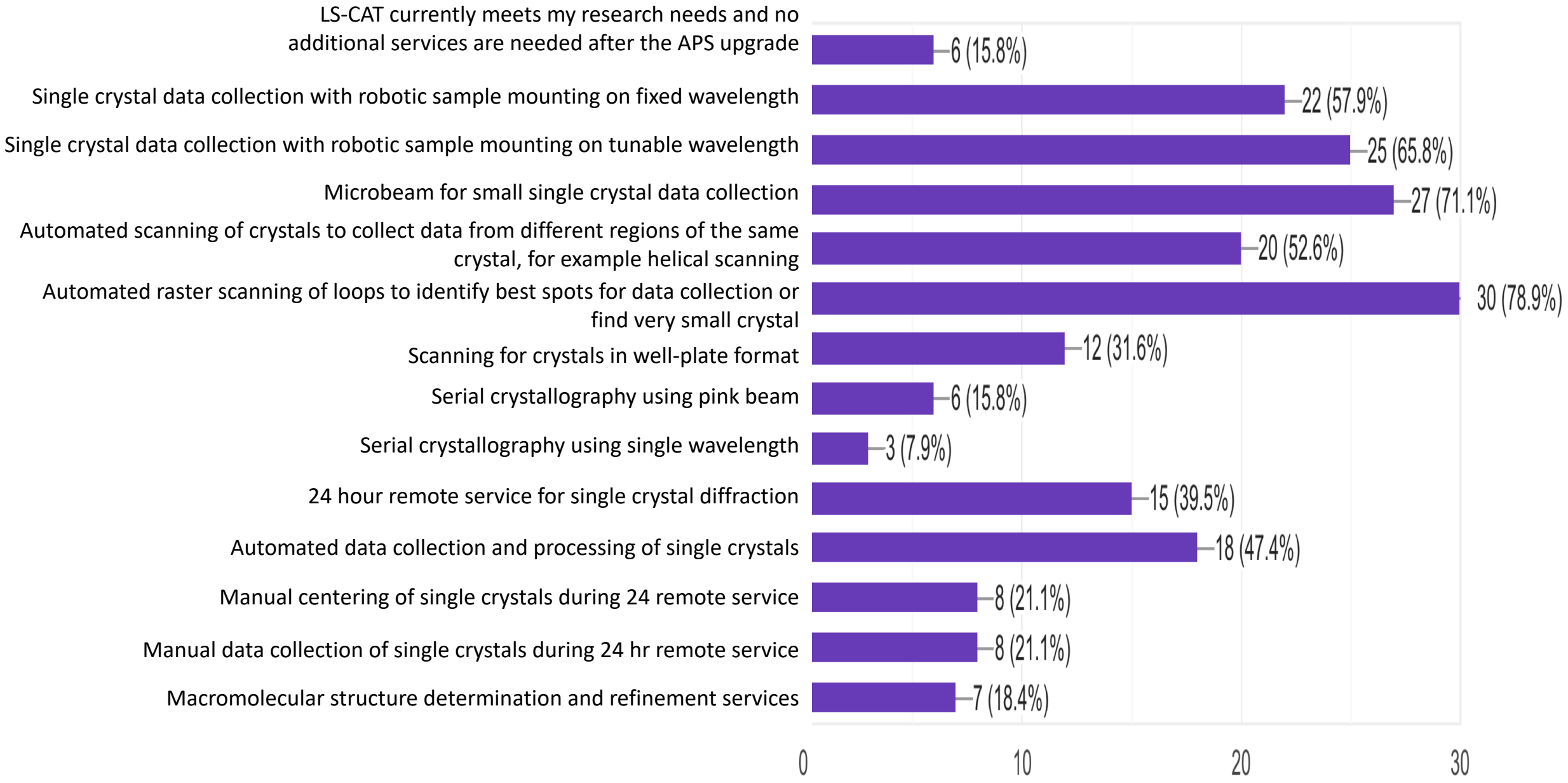
Response Rate/Institution



*University of Michigan: Crystallography Faculty, College of Engineering Faculty (5) (5 additional responses from graduate students/fellows)

Select all services desired at LS-CAT:

38 responses

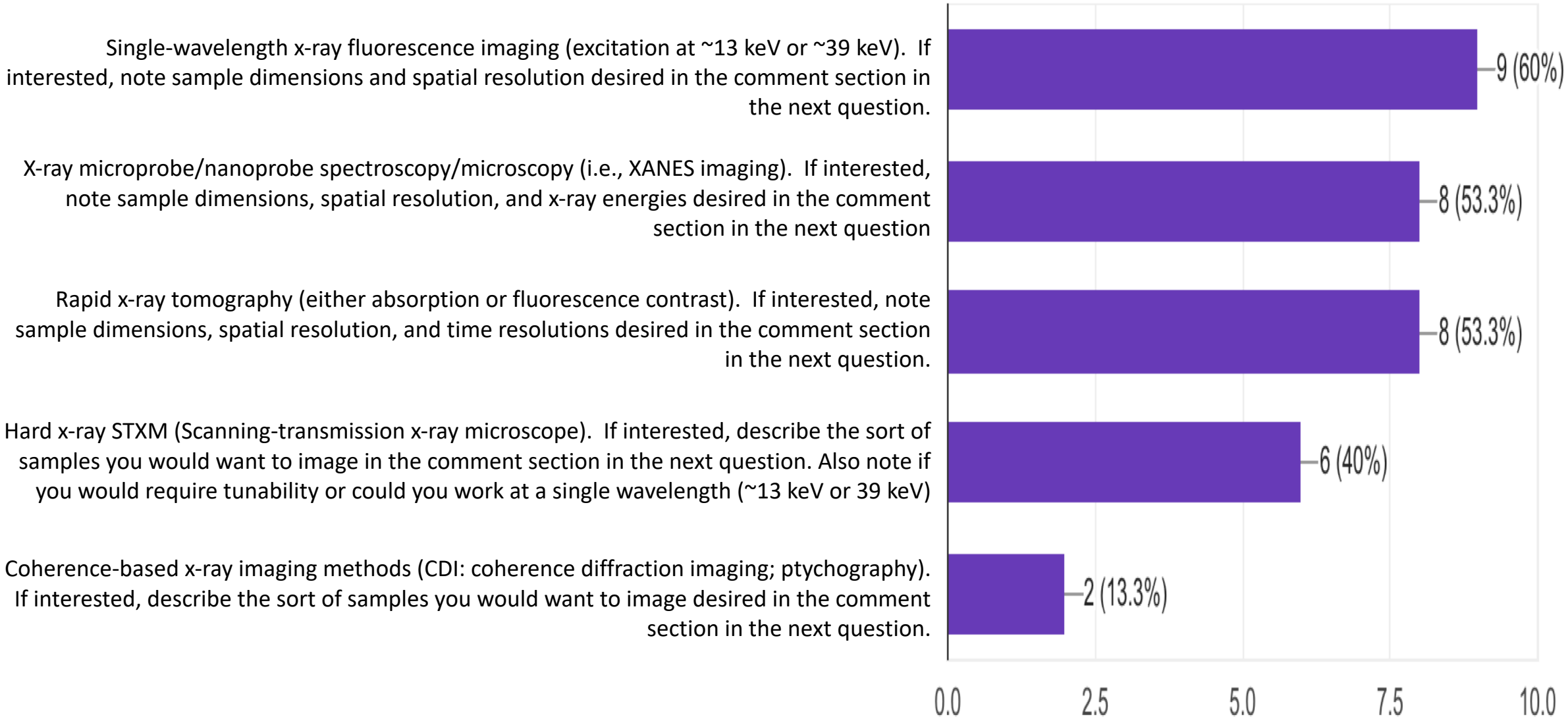


Please comment or suggest new crystallographic technologies/services not listed above that are desired here. Additionally, please comment if there are technologies/services which you specifically would lower the priority of (e.g. spend no or few resources on):

6 responses

For the most part my current needs are being met quite well so I don't really need, at this time, extra services.
Automatic rinsing of crystals with LN2
SAXS a huge need!
n/a
microspec
N/A

In addition to crystallographic measurements, the APS upgrade could (with appropriate investment) permit new capabilities at LS-CAT. Please check which capabilities would be desired: (15 responses)



Please comment or suggest new technologies/services (non crystallographic) not listed above that are desired, as well as notes from previous question here:

11 responses

Needed: 1) Automated estimation of radiation exposure for data collection strategies. 2) Increased dewar capacity (e.g. cassettes vs. pucks), especially if no puck swaps available during overnight remote.

the LS-CAT does not do scattering which is a key need. They only describe macromolecular crystallography which must be a small subset of faculty here at UM. if UM is investing only in crystallography, how is this investment supporting a larger fraction of UM? Please put in the detectors to allow more of the UM community to get beam access without having to go through the Users program.

A UV-Vis set up to record spectra prior, during and/or after X-Ray crystallography collection would be very useful for the study of metalloenzymes. This set up can be installed in any of the MX lines but it is availability in D would be most preferable.

For my applications I would like sample dimensions of ~ 1mm dia., spatial resolution of at least 1 um, and temporal resolution of ~1 s.

Radiation acoustic measurements for dosimetry and imaging

For x-ray fluorescence imaging, we are hoping to scan cells (tens of micrometer in diameter). Nanometer spacial resolution would be very ideal.

CryoEM/EM with ancillary equipment for macromolecules and cellular systems

bacteria or polysaccharide granules.